MCO-1D ORNITHOPTER

Physical Parameters Required Before Test Cards

Revision 18 — July 2026

Collect every item in this document before running tc01_start. Enter values into mco_build_measurements.py which writes /home/pi/knowledge/build_measurements.json. TC-01 reads this file at startup and uses actual values in the feasibility gate.

■ WARNING: If you skip any item in Section A (mass) or Section C (gear ratio), the TC-01 hover feasibility gate will classify the aircraft incorrectly.

■ WARNING: If you skip G4 (pitot valid), hover mode cannot engage. pitot_force_valid is available for bench testing without pitot installed.

● CAUTION: If you skip H4 (sector arc symmetry), a wing may hit the mechanical hard stop during the first frequency step. This can snap the CF rod.

A. MASS AND WEIGHT — CRITICAL FOR HOVER GATE

TC-01 step 16 computes max_thrust / MTOW. If MTOW is the design value (3.448 kg) but the aircraft actually weighs 5.1 kg, the gate classifies using a denominator that is 33% too low.

#	Item	Unit	How to measure	JSON key
A1	Total assembled mass (both batteries installed)	kg	Kitchen scale 5 kg / 1 g resolution	mtow_kg
A2	Mass with main battery removed	kg	Kitchen scale	mass_no_main_batt_kg
A3	Mass with both batteries removed	kg	Kitchen scale	mass_no_batts_kg
A4	Main LiPo mass (separate weigh)	g	Kitchen scale	main_lipo_g
A5	Servo LiPo mass (separate weigh)	g	Kitchen scale	servo_lipo_g

Design MTOW	3.448 kg (33.8 N)
Estimated actual	5.095 kg (49.9 N) — pending physical measurement
Hover at 5.095 kg	1.87 Hz — well within 3.0 Hz cap
Hover at 3.448 kg	1.54 Hz — also fine
Key point	Both weights hover comfortably. Measure real value for accurate TC-01 gate.

B. WING GEOMETRY — AFFECTS HOVER FREQUENCY PREDICTION

These are the values actually cut and built, not the design targets. Measure what exists.

#	Item	Unit	How to measure	JSON key
B1	W1 semi-span (pivot to rod tip)	mm	Steel rule, pivot centre to tip	span_w1_mm
B2	W2 semi-span	mm	Same	span_w2_mm
B3	W3 semi-span	mm	Same	span_w3_mm
B4	W4 semi-span	mm	Same	span_w4_mm
B5	Tip chord (at rod tip, before winglet)	mm	Callipers across chord	chord_tip_mm
B6	Root chord at bolt circle (maximum)	mm	Callipers	chord_root_mm
B7	Root taper length (bolt circle to tip chord)	mm	Steel rule	root_taper_mm
B8	Winglet span (rod tip to winglet tip)	mm	Steel rule	winglet_span_mm
B9	Winglet root chord	mm	Callipers	winglet_chord_root_mm
B10	Winglet tip chord	mm	Callipers	winglet_chord_tip_mm
B11	Winglet cant angle (from vertical)	°	Digital angle gauge	winglet_cant_deg

Target spans	W1=772, W2=749, W3=735, W4=712 mm
Target tip chord	165 mm
Target root chord	230-250 mm
Target root taper	180 mm
Target cant	75-80°

C. GEAR RATIO AND SECTOR ARC — CRITICAL

■ WARNING: C1 (gear ratio) affects every frequency calculation. If actual ratio differs from 7:1, the hover frequency prediction and vel_limit setting are wrong. Measure before TC-01.

■ WARNING: C2/C3 (sector arc symmetry) must be verified before any frequency above 0.5 Hz. Asymmetry >2° means the 4-degree mechanical reserve against the hard stop is one-sided. A wing hitting the hard stop at 2 Hz can snap the CF rod and destroy the load cell fixture.

#	Item	Unit	How to measure	JSON key
C1	Actual gear ratio (motor turns per sector revolution)	ratio	Command known motor counts via ODrive. Count sector degrees swept. 7 turns = 360° if 7:1.	gear_ratio
C2	Sector arc — positive side (mid-stroke to hard stop)	°	Digital angle gauge. Measure from neutral to each hard stop separately.	sector_arc_pos_deg
C3	Sector arc — negative side (mid-stroke to hard stop)	°	Same — measure other side	sector_arc_neg_deg
C4	Asymmetry (|C2 - C3|)	°	Computed. Must be <2°	sector_asymmetry_deg
C5	Software amplitude limit vs hard stop reserve	°	Hard stop - 48° must be ≥4° each side	sector_reserve_deg

✓ TC-01 step 05b (sector arc symmetry) automates this check. Run it before step 06 (motor spin). If any wing fails, re-seat the motor drum and re-verify.

D. SERVO RANGE AND FEEDBACK — REQUIRED FOR CALIBRATION

TC-01 steps 09 and 10 automate pitch and sweep calibration. These pre-checks confirm the servos can physically reach the required range.

#	Item	Unit	How to measure	JSON key
D1	Pitch servo neutral ADC count (0° AoA, all 8)	counts	Read MCP3208 at neutral command	pitch_adc_neutral_counts[8]
D2	Pitch servo +45° ADC count (all 8)	counts	Command +45°, read ADC	pitch_adc_pos45_counts[8]
D3	Pitch servo -8° ADC count (all 8)	counts	Command -8°, read ADC	pitch_adc_neg8_counts[8]
D4	Sweep servo neutral ADC count (60°, all 8)	counts	Read at neutral	sweep_adc_neutral_counts[8]
D5	Actual stow angle reached by each sweep servo	°	Command max aft. Measure with angle gauge. Min 85° required.	sweep_stow_deg[8]
D6	Sweep servo maximum forward angle	°	Command full forward, measure	sweep_fwd_deg[8]

■ WARNING: D5 is a gate: if any sweep servo cannot reach 85° aft, H-33 State C is mechanically unavailable on that wing. Inform the software: set_wing_stow_limit W1p 82 (for example).

✓ IB53BHU feedback output: 300 mV (0°) to 3000 mV (180°). MCP3208 VREF = 3.3V → ADC range 372-3723 counts for 0°-170°. Programme servo to 170° max travel via AGF-SPV3 programmer before installation.

E. PROPULSION — COLLECTED AUTOMATICALLY BY TC-01

These values are read by the S1 boards and HX711 load cell during TC-01. Listed here so you know what is being recorded.

#	Item	Unit	Source
E1	Motor baseline current at 5 rad/s (all 8)	A	Auto TC-01-05b
E2	Motor current at 0.5 Hz flapping (all 8)	A	Auto TC-01-07
E3	Motor current at 2.5 Hz flapping (all 8)	A	Auto TC-01-08
E4	FET temperature at 2.5 Hz × 60 s	°C	Auto TC-01-08
E5	Static thrust at each frequency step (N)	N	Auto TC-01-12
E6	Thermal ceiling frequency	Hz	Auto TC-01-11

F. CG AND STRUCTURAL

#	Item	Unit	How to measure	JSON key
F1	CG — fore-aft from W1 pivot	mm	Balance on known fulcrum	cg_fwd_mm
F2	CG — lateral (on centreline = 0)	mm	Balance	cg_lat_mm
F3	CG height above ground	mm	Steel rule	cg_height_mm
F4	Undercarriage height (root pivot to ground)	mm	Steel rule	h_gear_mm
F5	W4 wing tip to ground at full down-stroke	mm	Measure at 48° deflection — MUST BE >50 mm	tip_ground_clearance_mm
F6	Pitot tube tip offset from nose	mm	Steel rule	pitot_offset_mm

■ WARNING: F5 must be >50 mm. If less, reduce amplitude limit in wave library before TC-01. Wing strike on load cell fixture at 2 Hz = serious damage.

G. SENSOR VERIFICATION — BEFORE ANY POWER-ON

#	Item	Result	Command / Method
G1	DS18B20 OAT reading	°C	python3 -c "import glob; [print(open(f).read()) for f in glob.glob('/sys/bus/w1/devices/28*/temperature')]"
G2	IMU at I2C 0x68	pass/fail	i2cdetect -y 1 → must show 68
G3	Barometer at I2C 0x76	pass/fail	i2cdetect -y 1 → must show 76
G4	Pitot receiving MAVLink VFR_HUD	pass/fail	pitot_status on GS. Blow on tube — count must reach 3. Hover mode BLOCKED until this passes.
G5	All 8 CAN nodes responding	pass/fail	python3 mco_odrive_setup.py --verify → all 8 respond
G6	All 8 encoder magnets field_status=1	pass/fail	python3 mco_odrive_setup.py --check-magnet → all 8 = 1
G7	RC channels and lever direction	pass/fail	python3 mco_rc_verify.py → CH1=Roll, CH2=Pitch, CH3=Lever fwd=thrust, CH4=Yaw pedals

■ WARNING: G4 — Pitot valid: hover mode uses pitot airspeed to select control law. If pitot_valid is False, hover mode is permanently blocked (defaults to fixed-wing). For bench testing without pitot: pitot_force_valid on GS.

H. PHYSICAL CHECKS — BEFORE TC-01 FREQUENCY STEPS

#	Check	Method	Action if fail
H1	All 8 rope drives: no fraying, correct tension	Visual + tension check	Replace rope if any fraying
H2	All 16 servo horns secure	Pull test by hand	Re-secure with Loctite 243
H3	Load cell fixture rigid (no movement at 50 N push)	Push test by hand	Reinforce — must hold 115 N at TC-01 step 12
H4	Sector arc symmetry ≤2° (from C2/C3 measurements)	Angle gauge	Re-seat motor drum. Verify TC-01 step 05b before step 06.
H5	Wing tip ground clearance >50 mm at full stroke	Measure at 48° deflection	Reduce amplitude limit in wave library
H6	Covert flaps free to pivot, return to neutral	Deflect by hand	Free hinge / adjust spring
H7	Turbulator tape bonded, no bubbles	Visual	Re-apply lifting sections
H8	All 8 winglets aligned 75-80° cant, bonded secure	Angle gauge at winglet	Re-bond sleeve if any loose

QUICK REFERENCE — DESIGN vs ESTIMATED vs ACTUAL

Parameter	Design	Estimated	Actual (fill in)
Total mass	3.448 kg	5.095 kg	_____ kg
Wing loading	34.5 N/m²	45.7 N/m²	_____ N/m²
Hover frequency	1.54 Hz	1.87 Hz	_____ Hz (from TC-02)
Max thrust @ 3.0 Hz	78.4 N	128.8 N	_____ N (from TC-01)
Thrust / MTOW @ 3.0 Hz	2.32×	2.58×	_____ ×
Gear ratio	7:1	7:1 assumed	_____ :1 (measure C1)
Sector arc — positive side	50°	50°	_____ ° (measure C2)
Sector arc — negative side	50°	50°	_____ ° (measure C3)
Sector asymmetry	<2°	<2° assumed	_____ ° (C3-C2)
Sweep stow angle — each wing	85-95°	85°+	_____ ° (measure D5)
Pitot valid at boot	required	required	YES / NO (check G4)
CG fore-aft from W1 pivot	TBD	TBD	_____ mm (measure F1)
W4 tip clearance at 48°	>50 mm	>50 mm	_____ mm (measure F5)
Main LiPo mass	~420 g	420 g est	_____ g (weigh A4)
ADC counts at 0° AoA (W1p)	372	372	_____ counts (D1)
ADC counts at 170° (W1p)	3723	3723	_____ counts (D2)

✓ Fill in the Actual column from physical measurements before tc01_start. Enter into mco_build_measurements.py → /home/pi/knowledge/build_measurements.json.

✓ TC-01 reads build_measurements.json at startup. If file is missing: warning printed, design MTOW used.

WHAT HAPPENS IF YOU SKIP EACH ITEM

Skipped	Consequence
A1 actual mass	TC-01 gate uses 3.448 kg — may classify FREE_HOVER when overweight
C1 gear ratio	Hover frequency wrong — descent controller miscalibrated at TC-02
C2/C3 sector arc	Wing hits hard stop at frequency — CF rod failure or rope snap
D5 stow angle	H-33 State C commanded on wing that cannot reach stow — servo stall
F5 tip clearance	Wing strikes load cell fixture at first frequency step
G4 pitot valid	Hover mode blocked — cannot perform TC-04 FREE_HOVER profile
G6 magnet check	Cold-start position wrong — S1 sets wrong frequency
H4 sector symmetry	Duplicate of C2/C3 physical check — must confirm before electrical